**Project Title**

Recommendation Engine

*Project Report submitted for the partial fulfilment of grade for the course*

**Data Structures and Algorithms -3 (25CS2103E)**

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# ABSTRACT

This project aims to develop an efficient Recommendation Engine that provides personalized recommendations to users based on their preferences, interests, and interaction history. In today's digital platforms, users are presented with a large number of products, movies, books, or other content, making it difficult to find relevant items quickly. The proposed system addresses this problem by analyzing user preferences and item characteristics to generate a ranked list of suitable recommendations.

The system collects and processes user interaction data such as previously selected, viewed, searched, or rated items. It calculates the relevance of each item based on user interests and content features and then ranks the items according to their relevance. Various Data Structures and Algorithms are used to improve the efficiency and performance of the recommendation process. Arrays, strings, lists, and dynamic data structures are used for storing and managing user and item information. Searching and sorting algorithms are applied for fast retrieval and ranking of recommendations, while string-searching techniques are used to efficiently match user interests with item information.

The system is designed to provide relevant, diverse, and limited recommendations according to the user's requirements. It also focuses on reducing unnecessary searching and improving the overall response time. The project demonstrates the practical application of Data Structures and Algorithms-3 concepts in developing an efficient and scalable recommendation system. The proposed Recommendation Engine can be further extended for real-world applications such as e-commerce, entertainment platforms, online learning systems, and content recommendation services.

Signature of the Guide